

Risk Management

Overdrive

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Summary

OverDrive – Risk Management

1- Risk analysis (Matrice)

2- Mitigation strategy

1- Technical risks

- 1- Dependence on data sources
- 2- VR performance and stability
- 3- Live synchronization

2- Legal risks

- 1- Distribution rights and proprietary data

3- Financial risks

3- Green IT & Societal Impact

1- Environmental impact

2- Societal impact

3- Material

4- Resilience and continuity of service

1- Backups (Backups)

2- High availability

3- Maintenance & updates

5- Conclusion

Introduction

The objective of risk management is to identify the main threats that may impact the smooth running of the OverDrive project, then to propose appropriate preventive measures to limit their impact.

OverDrive is a project with a strong technological component, combining :

- Real-time sports data,
- a multi-client backend (VR, mobile, web),
- an immersive VR/AR experience,
- a reliance on external sources (providers).

Each risk is assessed according to :

- Probability: Low / Medium / High
- Impact: Low / Medium / High / Very High
- Overall Level: Moderate / High / Critical

1- Risk analysis (Matrice)

1- Synthetic table

Hazard	Likelihood	Impact	Level
Dependence on external APIs	High	High	Criticism
VR performance issues	Average	High	High
Live Sync Shift	Average	Average	Moderate
Legal problem (media / data rights)	Low (personal use)	Very high	Criticism
Exceeding the cloud budget	Low	Average	Moderate
Organizational delay	Average	Average	Moderate

Matrix reading

Critical risks are those combining :

- a very significant impact on the project,

- and/or a high probability of occurrence.

In OverDrive, two risks stand out in particular. :

1. Dependence on external APIs
2. Legal risks related to broadcasting rights

These risks shape the architectural choices of the project.

2- Mitigation strategy

The strategy is based on a double approach :

- **Preventive** : Reduce the likelihood of occurrence
- **Curative** : Reduce the impact in case of risk realization

1- Technical risks

Dependence on data sources

Problem :

The data comes from external APIs that may change, be restricted, or become unavailable.

Preventive strategy

- Modular architecture per provider
- Independent internal normalization (OverDrive Schema)
- Ability to dynamically disable a provider

Curative strategy

- Temporarily switch to **replay mode**
- Display a transparent message to users
- Quickly replace the source with a compatible alternative

VR performance and stability

Problem :

Drop in framerate, latency or user discomfort in immersive environments.

Preventive

- Unity scene optimization
- Regular testing on target hardware (Meta Quest)

- Limiting the number of simultaneous widgets
- Reducing unnecessary polling

Curative

- Enabling a "Low Performance" mode
- Dynamic reduction of animations
- Disabling secondary visual effects

Live synchronization

Problem :

Offset between video broadcast and displayed data.

Preventive

- Timestamp-based system on the backend
- Configurable offset management

Curative

- Manual recalibration by the user
- Server-side offset correction

2- Legal risks

Distribution rights and proprietary data

Problem :

Indirect use of data from competitions subject to commercial rights.

Preventive

- **Data-first** positioning, with no redistribution of video streams
- No hosting of copyrighted content
- Legal audit if transitioning to the commercial phase

Curative

- Immediate deactivation of a non-compliant provider
- Rapid technical compliance
- Adaptation of the business model if necessary

3- Financial risks

Exceeding cloud costs

Problem :

Unexpected increase in costs related to the increase in load.

Preventive

- VPS for MVPs (no initial oversizing)
- Cloud cost monitoring
- Monthly budget forecast
- Automatic alerts in case of overspending

3- Green IT & Societal Impact

OverDrive integrates a reflection of digital sobriety and responsibility.

1- Environmental impact

Accommodation

Cloud servers consume energy.

Reduction measures :

- Sizing adapted to the actual load
- Redis cache to limit recalculations
- Stream compression
- Reduction of unnecessary requests

Real-time data volume

Real-time increases network traffic.

Measures :

- WebSocket message frequency limitation
- Only send necessary data
- Configurable "Low Data" mode

Material

VR and high-performance PCs have an environmental cost.

Measures :

- GPU optimization to limit consumption
- Lengthening the material life
- Limitation of frequent equipment renewal

2- Societal impact

Accessibility

Risk: exclusion of people who do not have a VR headset.

Answer :

- Companion mobile app
- Web dashboard
- Simplified, non-immersive mode

Ethics & digital use

Risk: overconsumption of screens, dependence on real-time information.

Answer :

- No addictive mechanisms
- No excessive notifications
- An information-focused experience, not gamification

4- Resilience and continuity of service

1- Backups

- Automatic daily backup of PostgreSQL
- Backup storage in separate space
- Multi-day retention

In case of incident :

- Fast food possible
- Limited data loss

2- High availability

For the MVP

- Basic VPS
- Manual restart accepted
- No multi-region

For a future evolution

- Load balancing

- Database replication
- Auto-scaling backend
- Proactive monitoring

3- Maintenance & updates

- Regular update of dependencies
- Security scan (npm audit or equivalent)
- Testing before production
- API versioning (ex: /v1)

General conclusion

The OverDrive project presents clearly identified technical, legal, organizational, financial and environmental risks.

For each major risk :

- A preventative strategy reduces the likelihood of occurrence.
- A curative strategy reduces the impact if an event does occur.

The project also includes :

- A digital sobriety approach (Green IT)
- A reflection on accessibility and ethical use
- A scalable architecture ensuring resilience

This structured approach to Risk Management strengthens the technical, strategic and responsible credibility of OverDrive, in line with a sustainable and scalable product ambition.